

Enamundram Naga Karthik

Postdoctoral Researcher, University Health Network  **UHN** Canada's Hospital

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 emvnagakarthik  [naga-karthik.github.io](https://github.com/naga-karthik)  [enamundram-naga-karthik](https://www.linkedin.com/in/enamundram-naga-karthik)  [naga-karthik](https://twitter.com/naga-karthik)  scholar

RESEARCH INTERESTS

Multimodal Learning, Tabular Foundation Models, Survival Analysis, Lifelong Learning, Computer Vision, Medical Image Analysis, Generative Models

RESEARCH OBJECTIVE

Postdoctoral fellow with experience in processing large-scale medical datasets, developing deep learning techniques for medical image analysis, and contributing to open-source software. Interested in collaborating with research and clinical teams developing multimodal predictive models for precision medicine.

EDUCATION

- **Polytechnique Montréal / Mila - Québec AI Institute** 09/2021 - 12/2025
Ph.D. in Biomedical Engineering Montréal, Canada
 - Thesis: Towards automatic spinal cord MRI analysis for improved estimation of imaging biomarkers [[thesis](#)] ("Excellent" mention)
 - Supervisor and Co-Supervisor: [Prof. Julien Cohen-Adad](#) and [Prof. Sarath Chandar](#)
 - GPA: 4.0/4.0
- **École de Technologie Supérieure** 09/2019 - 07/2021
Master of Applied Sciences (M.A.Sc) in Electrical Engineering Montréal, Canada
 - Thesis: Unsupervised Three-dimensional Segmentation of Scoliotic Spines from MR Volumes with Uncertainty Estimation [[thesis](#)]
 - Supervisor and Co-Supervisor: [Prof. Catherine Laporte](#) and [Prof. Farida Cheriet](#)
 - Grade: 4.3/4.3
- **Shiv Nadar University** 07/2015 - 05/2019
Bachelor of Technology in Electronics and Communication Engineering with Minor in Mathematics Greater Noida, India
 - Thesis: Extraction and Visualization of the Features of Fingerprints using Conventional Methods and CNNs
 - Supervisor: Prof. Madan Gopal
 - GPA: 9.19/10

WORK EXPERIENCE

- **University Health Network** 01/2026 - Current
Canada Leads Postdoctoral Fellow Toronto, Canada
 - Project: Accurate prioritization of patients on the liver transplant waitlist using tabular foundation models.
 - Principal Investigators: [Dr. Mamatha Bhat](#) and [Prof. Rahul G. Krishnan](#).

PUBLICATIONS

B=BOOK CHAPTER, C=CONFERENCE, J=JOURNAL, P=PREPRINT

- [C.5] **Dynamic Robot-Assisted Surgery with Hierarchical Class-Incremental Semantic Segmentation**
J. Hindel, E. Mekic, [E. N. Karthik](#), R. Mohan, D. Cattaneo, M. Kalweit, A. Valada
MICCAI Workshop on Applications of Medical Artificial Intelligence [[paper](#)][[code](#)] [MICCAI AMAI '25]
- [P.2] **Monitoring morphometric drift in lifelong learning segmentation of the spinal cord**
[E. N. Karthik](#), S. Bédard, J. Valošek, ..., 50 more authors, ..., S. Chandar, J. Cohen-Adad
Imaging Neuroscience (formerly NeuroImage) [[paper](#)][[code](#)] [ImagNeuro '26]
Accepted as a Poster at the Conference on Lifelong Learning Agents Workshop Track [CoLLAs '25]
- [J.6] **Automatic segmentation of spinal cord lesions in MS: A robust tool for axial T2-weighted MRI scans**
[E. N. Karthik*](#), J. McGinnis*, R. Wurm, S. Ruehling, R. Graf, J. Valosek, P.L. Benveniste, M. Lauerer, J. Talbott, R. Bakshi, S. Tauhid, T. Shepherd, A. Berthele, C. Zimmer, B. Hemmer, D. Rueckert, B. Wiestler, J. S. Kirschke, J. Cohen-Adad, Mark Muehlau
Imaging Neuroscience (formerly NeuroImage) [[paper](#)][[code](#)] [ImagNeuro '25]
- [J.5] **Towards Contrast-Agnostic Segmentation of the Spinal Cord**
S. Bédard*, [E. N. Karthik*](#), C. Tsagkas, E. Pravatà, C. Granziera, A. C. Smith, K. A. Weber II, J. Cohen-Adad
Medical Image Analysis (2025 Impact Factor: 11.8) [[paper](#)][[code](#)] [MedIA '25]

- [J.4] **SCIseg: Automatic Segmentation of Intramedullary Lesions in Spinal Cord Injury**
E. N. Karthik*, J. Valošek*, A. C. Smith, D. Pfyffer, S. Schading-Sassenhausen, L. Farner, K. A. Weber II, P. Freund, and J. Cohen-Adad
Radiology: Artificial Intelligence (2025 Impact Factor: 13.2) [paper][code] [RadAI '24]
- [C.4] **SCIsegV2: A Universal Tool for Segmentation of Intramedullary Lesions in Spinal Cord Injury**
E. N. Karthik*, J. Valošek*, L. Farner, D. Pfyffer, S. Schading-Sassenhausen, A. Lebet, G. David, A. C. Smith, K. A. Weber II, M. Seif, P. Freund, J. Cohen-Adad
MICCAI Workshop on Applications of Medical Artificial Intelligence [paper][code] [MICCAI AMAI '24]
- [C.3] **Contrast-agnostic Spinal Cord Segmentation: A Comparative Study of ConvNets and Vision Transformers**
E. N. Karthik, S. Bédard, J. Valošek, S. Chandar, J. Cohen-Adad
Medical Imaging with Deep Learning: Short-paper Track [paper][code] [MIDL Short '24]
- [J.3] **Uncertainty Estimation in Unsupervised MR-CT Synthesis of Scoliotic Spines**
E. N. Karthik, F. Cheriet and C. Laporte
IEEE Open Journal of Engineering in Medicine and Biology (Impact Factor: 2.7) [paper][code] [IEEE OJEMB '23]
- [C.2] **Segmentation of Multiple Sclerosis Lesions across Hospitals: Learn Continually or Train from Scratch?**
E. N. Karthik, A. Kerbrat, P. Labauge, T. Granberg, J. Talbott, D. Reich, M. Filippi, R. Bakshi, V. Callot, S. Chandar, J. Cohen-Adad
Medical Imaging Meets NeurIPS Workshop [paper][code] [MedNeurIPS '22]
- [J.2] **Label fusion and training methods for reliable representation of inter-rater uncertainty**
A. Lemay, C. Gros, E.N. Karthik, J. Cohen-Adad
Machine Learning for Biomedical Imaging [paper][code] [MELBA '22]
- [P.1] **Team NeuroPoly: Pipelines for the MICCAI 2021 MS New Lesions Segmentation Challenge**
U. Macar, E. N. Karthik, C. Gros, A. Lemay, J. Cohen-Adad
MICCAI 2021 Challenge: Multiple Sclerosis New Lesions Segmentation [paper][code] [MICCAI '21]
- [C.1] **3D segmentation of the scoliotic spine from MRI using unsupervised volume-based MR-CT synthesis**
E. N. Karthik, F. Cheriet and C. Laporte
SPIE: Medical Imaging [paper][code] [SPIE '21]
- [B.1] **Extraction of the Features of Fingerprints Using Conventional Methods and CNNs**
E. N. Karthik, G. Madan
Machine Learning Algorithms and Applications [paper] [Wiley '21]
- [J.1] **Automatic tongue surface extraction from three-dimensional ultrasound vocal tract images**
E. N. Karthik, E. Karimi, S. M. Lulich, C. Laporte
Journal of the Acoustical Society of America [paper] [JASA '20]

HONORS AND AWARDS

- **UHN Ajmera Transplant Centre Research Institute Best Poster Award** 2026
University Health Network (affiliated with the University of Toronto) [award]
◦ Recipient of the best poster award in the [Fifth Annual Ajmera Transplant Centre Research Day](#).
- **UHN Canada Leads Postdoctoral Fellowship | 150,000 CAD** 2025
University Health Network (affiliated with the University of Toronto)
◦ Recipient of the competitive postdoctoral fellowship funded by the [Canada Leads initiative](#) to recruit 100 world-leading early career scientists.
- **QBIN Travel for Training Awards | 3,000 CAD** 2024
Quebec Bio-Imaging Network / Réseau de Bio-Imagerie du Québec [letter]
◦ Recipient of the travel award for a research visit to Technische Universität München, Germany.
- **DAAD Short-Term Research Grants | 7,800 EUR** 2024
German Academic Exchange Service / Deutscher Akademischer Austauschdienst [letter]
◦ Recipient of the prestigious grant for a 6-month research visit to Technische Universität München, Germany.
- **UNIQUE Excellence Doctoral Scholarship | 15,000 CAD** 2022
Unifying Neuroscience and Artificial Intelligence - Québec / Union Neurosciences et Intelligence Artificielle - Québec [letter]
◦ Recipient of the doctoral scholarship from UNIQUE, an FRQNT-funded research center.
- **FRQNT Doctoral Research Scholarship | 84,000 CAD** 2021
Fonds de recherche du Québec Nature et technologies [letter]

- Secured funding for 4 years from the prestigious doctoral scholarship by ranking 1st out of 29 applicants.
- **Governor General of Canada's Academic Gold Medal** 2021
Government of Canada [letter]
 - Awarded Canada's most prestigious academic medal for highest CGPA during my master's studies at ÉTS.
- **Best Master's Thesis Award | 1,000 CAD** 2021
École de Technologie Supérieure Montréal [letter]
 - Recognized as the best Master thesis upon the recommendation of the jury for an "Excellent" thesis defense
- **Master's Tuition Fee Exemptions | 13,000 CAD** 2020
École de Technologie Supérieure Montréal
 - Received tuition fee exemptions for 3 consecutive semesters – Winter, Summer, Fall 2020.
- **ÉTS Master's Internal Scholarship | 5,000 CAD** 2020
École de Technologie Supérieure Montréal
 - Received the scholarship based on academic excellence and quality of the master's research project.
- **ÉTS Research Internship Travel and Subsistence Allowance | 4,200 CAD** 2018
École de Technologie Supérieure Montréal
 - Received travel and subsistence allowance for my undergraduate research internship at LATIS Lab, ÉTS Montreal.

RESEARCH EXPERIENCE

- **TUM-NIC and AIM Lab, Technical University of Munich [website]** 07/2024 - 11/2024
Visiting PhD Researcher Munich, Germany
 - Visited Prof. [Daniel Rueckert](#)'s Lab for AI in Medicine and Prof. [Mark Muehlau](#)'s Lab at TUM Neuroimaging Center on a 4-month DAAD-funded research stay.
 - Investigated the impact of preprocessing techniques, namely, *stitching* and *straightening* of individually-acquired chunks of the spinal cord towards multiple sclerosis lesion segmentation in spinal cord MRI.
- **NeuroPoly Lab, Polytechnique Montreal [website]** 06/2021 - 08/2021
Graduate Research Intern Montreal, Canada
 - **Automatic Segmentation of New Multiple Sclerosis Lesions:** Worked on developing a deep learning-based method for the soft segmentation of multiple sclerosis lesions as part of the MICCAI 2021 MS New Lesions Segmentation Challenge.
 - Our team secured 2nd position out of 23 teams on one of the challengers' evaluation metrics.
- **LATIS Lab, Ecole de Technologie Supérieure, Montreal [website]** 05/2018 - 08/2018
Undergraduate Research Intern Montreal, Canada
 - Applied the concepts of 3D phase symmetry and active surfaces to detect and extract the surface of the tongue from raw 3D ultrasound (US) images under the supervision of Prof. Laporte at LATIS.
 - Work done in collaboration with the Speech and Hearing Sciences Department, Indiana University, Bloomington.
- **Computer Science and Engineering Department, IIT Bombay** 05/2017 - 07/2017
Undergraduate Research Intern Mumbai, India
 - Adapted the concepts of filtered back projection (FBP), dictionary learning and conjugate gradient methods to reconstruct computed tomography (CT) images of a walnut.
 - Worked under the supervision of Prof. [Ajit Rajwade](#) at the Department of Computer Science.

OPEN-SOURCE CONTRIBUTIONS

- **Spinal Cord Toolbox (SCT) | Command-line tools for the analysis of spinal cord MRI data** 09/2022 - 12/2022
Maintained by: NeuroPoly Lab [github]
 - Integrated into SCT the pathology-specific segmentation models I have developed for clinical collaborators.
 - Examples: Spinal cord injury lesions [\[paper\]](#)[\[usage\]](#), contrast-agnostic segmentation [\[paper\]](#)[\[usage\]](#), multiple sclerosis lesions [\[paper\]](#)[\[usage\]](#)
- **MetricsReloaded | New Framework for Biomedical Image Analysis Validation** 09/2022 - 12/2022
Maintained by: Project MONAI [github]
 - Added relative volume error and lesion-wise metrics (sensitivity, F₁-score) for better segmentation evaluation.
- **ivadomed | Open-source package for Medical Image Analysis with Deep Learning** 09/2022 - 12/2022
Maintained by: NeuroPoly Lab [github]
 - Contributed by adding new features and models developed by the lab.

REVIEWING

- **Journals:** [Computers in Biology and Medicine](#), [Neural Networks](#), [Quantitative Imaging in Medicine and Surgery](#) (Reviewer of the month award), [Lancet Digital Health](#), [Liver Transplantation](#)
- **Conferences:** [MICCAI 2025](#) (Outstanding Reviewer: Honorable Mention), [MLHC 2026](#), [ICML 2026](#) – [Foundation Model for Structured Data \(FMSD\) Workshop](#)

TEACHING EXPERIENCE

- **INF8245E - Machine Learning | Course Instructor: [Sarath Chandar](#)** 09/2022 - 12/2022
Polytechnique Montréal | Teaching Assistant [\[website\]](#)
 - Responsible for designing a coding assignment on introductory ML concepts including k-Nearest Neighbours, Bayes Classifier and Logistic Regression.
 - Holding weekly office hours and assisting the instructor in grading assignments.
- **IFT6135 - Learning Representations | Course Instructor: [Aaron Courville](#)** 01/2022 - 04/2022
Université de Montréal | Teaching Assistant [\[website\]](#)
 - Responsible for designing assignments on deep generative models and self-supervised learning.
 - Holding weekly office hours and assisting the instructor in grading assignments.
- **Deep Learning Summer School** 01/2022 - 04/2022
Neuromatch Academy | Teaching Assistant [\[website\]](#)
 - Responsible for teaching a 3-week course on deep learning to a group of 10 graduate students.
 - Topics ranged from basic (Multilayer Perceptrons, Optimization) to advanced (Reinforcement, Continual, and Self-supervised Learning). Complete syllabus can be found [here](#).

MENTORSHIP

- **Graduate Mentor, Industry Mentorship Program** 05/2024 - 12/2024
Mila - Québec AI Institute
 - Responsible for mentoring two graduate students, Jeremy Kaufman and Guillaume Charron, enrolled in Mila's Professional Master's program during their industrial internships at the [National Film Board of Canada](#) (NFB).
 - * **Project:** Automatic indexing and retrieval of video data. Objectives of their internship were to: (i) create an indexing tool with automatic captioning, video tagging and synopsis generation on NFB's decades-long archived data, and (ii) using multi-modal (text, audio, video) models for automatic content retrieval based on user queries.
 - * **Responsibilities:** My tasks involved assisting the students in their transition from academia to industry, ensuring progress in the company's projects through weekly meetings, providing technical guidance by discussing papers, solving bugs, etc.

VOLUNTEER EXPERIENCE

- **Supervision Requests Evaluator** 11/2024 - 12/2024
Mila - Québec AI Institute [\[website\]](#)
 - Evaluated 14 Masters and PhD graduate school applications to Mila
 - Responsibilities included evaluating the applicants' research potential and their fit within various research groups led by Mila professors through their grades, CVs, statements-of-purpose and recommendation letters.
- **Graduate School Application Assistance for Underrepresented Students in AI** 2023-current
Chandar Research Lab Initiative [\[website\]](#)
 - Assisted 8 students from under-represented countries for their graduate school applications by providing them tips to improve their CV and statements-of-purpose.
- **Organizing Committee Member for the Conference on Lifelong Learning Agents (CoLLAs)** 08/2022
Polytechnique Montréal / Mila - Québec AI Institute [\[website\]](#)
 - Member of the organizing committee for the first international conference on lifelong learning agents.
 - Responsible for back-end development and maintenance, mainly in hosting the conference website on Amazon Web Services (AWS) and handling accounting/registrations via Stripe.
- **Student Reviewer at the Service d'Aide à la Rédaction d'Articles (SARA)** 10/2019 - 06/2021
École de Technologie Supérieure, Montréal [\[website\]](#)
 - SARA is a scientific community at ÉTS aiming to help graduate researchers better the art of academic writing.
 - Volunteered for two paper-review requests, which involved the tasks of critically analyzing the authors' papers and providing insightful comments on the clarity and expression of their work.

TECHNICAL SKILLS

- **Programming and Markup Languages:** Python, Bash, L^AT_EX, Markdown, HTML, Git, Slurm
- **Data Science & Machine Learning:** PyTorch, scikit-learn, pandas, nibabel, matplotlib
- **Cloud Services:** AWS S3, AWS CloudFront
- **Specialized Software:** 3DSlicer, FSLeves, ITKSnap

ADDITIONAL INFORMATION

Languages: Hindi, Telugu (Native), English (C2 Level), French (Intermediate)

Sports: Winner of the Shiv Nadar University (SNU) Table Tennis Championship (2018, 2019). Winner of the best player award in SNU Table Tennis Championship for the academic year 2017-18.

Interests/Hobbies: Finisher of 5 Half Marathons (21.1 km). Best finish time 01:49:57.